

# Youngmin Kim

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## Professional Summary

*A Ph.D. Student in Artificial Intelligence with a research focus on **video understanding** and **human-interacted AI**. My work investigates how AI systems can interpret complex visual information from videos and connect this understanding to human perception and communication, with particular interest in **nonverbal expressions** and **sign language**.*

## Education

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|------------------------------------------|-----------------------|
| <b>Yonsei University</b>                 | Seoul, South Korea    |
| Ph.D. Student in Artificial Intelligence | Sep. 2023 – Present   |
| <b>Incheon National University</b>       | Incheon, South Korea  |
| B.S. in Economics & Computer Science     | Mar. 2016 – Mar. 2023 |

## Publications (\*: Equal Contribution)

### Conference Publications

1. **ViKey: Enhancing Temporal Understanding in Videos via Visual Prompting** [paper]  
Yeonkyung Lee\*, Dayun Ju\*, **Youngmin Kim**, Seil Kang, Seongjae Hwang  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
2. **MAVL: A Multilingual Audio-Video Lyrics Dataset for Animated Song Translation** [paper]  
Woohyun Cho, **Youngmin Kim**, Sunghyun Lee, Youngjae Yu  
Empirical Methods in Natural Language Processing (EMNLP), 2025
3. **Speaking Beyond Language: A Large-Scale Multimodal Dataset for Learning Nonverbal Cues from Video-Grounded Dialogues** [paper]  
**Youngmin Kim**\*, Jiwan Chung\*, Jisoo Kim, Sunghyun Lee, Sangkyu Lee, Junheok Kim, Cheoljong Yang, Youngjae Yu  
Association for Computational Linguistics (ACL), 2025
4. **Scalp Diagnostic System With Label-Free Segmentation and Training-Free Image Translation** [paper]  
**Youngmin Kim**\*, Saejin Kim\*, Hyeon Moon, Youngjae Yu, Junhyug Noh  
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025
5. **A 2-Stage Model for Vehicle Class and Orientation Detection with Photo-Realistic Image Generation** [paper]  
**Youngmin Kim**\*, Donghwa Kang\*, Hyeongboo Baek  
IEEE International Conference on Big Data (IEEE BigData), 2022
6. **Pedestrian Accident Prevention Model Using Deep Learning and Optical Flow** [paper]  
**Youngmin Kim**\*, Donghwa Kang\*, Hyeongboo Baek  
Korea Computer Congress (KCC), 2021  
**Best Paper Award (Undergraduate Session)**

### Journal Publications

1. **AI-Powered Real-Time System for Automated Concrete Slump Prediction via Video Analysis** [paper]  
**Youngmin Kim**, Giyeong Oh, Kwangsoo Youm\*, Youngjae Yu\*  
Automation in Construction (AiC), 2026, IF: 11.5
2. **Preprocessing for Keypoint based Sign Language Translation without Glosses** [paper]  
**Youngmin Kim**, Hyeongboo Baek.  
Sensors, 2023, IF: 3.8
3. **A Study of Tram-Pedestrian Collision Prediction Method Using YOLOv5 and Motion Vector** [paper]  
**Youngmin Kim**, Hyeonuk Ahn, Heegyun Jeon, Jinpyung Kim, Gyujin Jang, Hyeonchyeol Hwang

Korea Information Processing Society (KIPS), 2021

4. **Optical Flow Estimation Techniques and Recent Research Trends** [paper]  
Youngmin Kim, Hyeonuk Ahn, Jinpyung Kim  
Korea Information Processing Society Reviews, 2021

### Pre-prints

1. **Towards Continuous Sign Language Conversation from Isolated Signs** [paper]  
Youngmin Kim, Kyobin Choo, Jiwoo Park, Minseo Kim, Chanyoung Kim, Junhyeok Kim, Seongjae Hwang  
Pre-print
2. **QWERTY: Training-Free Motion Control via Query-Warped Video Diffusion Transformers** [paper]  
Kyobin Choo, Youngmin Kim, Hyunkyung Han, Geunrip Park, Chanyoung Kim, Sunyoung Jung, Seongjae Hwang  
Pre-print
3. **Sparse Feature Analysis and Control for Robust Vision-Language-Action Models** [paper]  
Minseo Kim, Junhyeok Kim, Jinyeong Kim, Youngmin Kim, Seongjae Hwang  
Pre-print
4. **Attention Weight as a Hierarchical Tree in ViT** [paper]  
Junhyeok Kim, Minseo Kim, Seil Kang, Youngmin Kim, Seongjae Hwang  
Pre-print

### Experience

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#### MIR Lab, Yonsei University

*Undergraduate Researcher* Mar. 2022 – Mar. 2023

#### RAISE Lab, Incheon National University

*Undergraduate Researcher* Dec. 2021 – Feb. 2023

#### CVAI Lab, Advanced Institute of Convergence Technology (AICT)

*Research Assistant* Sep. 2020 – Aug. 2021

### Scholarly Activities

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#### Invited Speaker

- **Samsung AI Education** 2023  
Introduction to Multimodal Research

#### Teaching Assistant

- Introduction to AI Research (Yonsei University) 2024
- Reinforcement Learning (Yonsei University) 2024
- Basic of PyTorch (Upstage) 2023

### Academic Service

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#### Journal Reviewer

- IEEE Transactions on Image Processing

### Selected Projects

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#### LG Electronic

*Multimodal Contents Recommendation System* Mar. 2025 – Aug. 2025

#### GS E&C

*Video-based Slump Prediction*

Dec. 2023 – Feb. 2025

**NCSOFT**

*Multimodal Conversational Agent*

Jan. 2024 – Jun. 2024

## Awards

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**AI-Hub Idea Challenge Competition** 2021

**1st prize**, hosted by South Korea's National Information Society Agency (NIA)

**KCC 2021 Undergraduate Paper Competition** 2021

**1st prize**, hosted by Korean Institute of Information Scientists and Engineers (KIISE)

**KED 2021 Industrial Innovation Big Data Platform Competition** 2021

**2nd prize**, hosted by Korea Enterprise Data (KED)

## Patents

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**Method and apparatus for avoiding collision between vehicle and object, computer program** 2023

KR102533419B1

**Electronic apparatus and method for searching distress, unmanned aerial vehicle, computer program** 2023

KR102587311B1

**Apparatus and method for analyzing data, apparatus and method for predicting abnormality, computer program** 2023

KR102480394B1